Inequalities — problem 7.56

A. Engel — Problem-Solving Strategies · 1998

For any triangle with sides a, b, c,

$$a^2b(a-b) + b^2c(b-c) + c^2a(c-a) \geq 0.$$

(Proposed by Klamkin and used in the IMO 1983. Due originally to E. Catalan, *Educational Times* N.S. 10, 57 (1906). The source is cited in [3].)